

The Cubic Boundary for Restricted Partitions

Sharp volume laws, uniform all-order expansions,
exponentially small corrections, inverse asymptotics,
and a Poisson law for repeated parts

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Abstract

Let $p_m(N)$ count partitions of N into parts at most m , equivalently partitions into at most m parts. We give an elementary necessary and sufficient condition for the volume approximation:

$$p_m(N) \sim \frac{N^{m-1}}{m!(m-1)!} \iff \frac{N}{m^3} \longrightarrow \infty, \quad m \longrightarrow \infty.$$

This proves, and sharpens, a condition still described as conjectural in OEIS A238016 and A258670. The sufficiency has classical antecedents and is not claimed as a new discovery. At the critical scale $N \sim cm^3$, the ratio to volume tends to $e^{1/(4c)}$. We prove a centered expansion uniform for $N \geq c_0m^3$, give an all-order coefficient algorithm and five explicit corrections for A238608, and derive a second expansion that resolves actual exponential sectors for $N = q^m$. Both types are inverted, with careful separation of smooth asymptotic charts from integer-valued inverse functions. In the conjugate model with at most m parts, the number of equal adjacent parts converges to $\text{Poisson}(1/(2c))$; its probability generating function has an explicit first finite-size correction. Exact computation checks 258 samples, including all displayed initial A238608 values, and the package supplies symbolic derivation and numerical verification code.

Status. This is a self-contained proof-and-extension report. Its leading partition formulas and first-wave machinery have substantial classical precedents. The explicit refinements below are derived here, but no exhaustive priority claim is made. The proofs have not been independently peer reviewed or formalized in Lean. Exact finite checks corroborate formulas; they do not replace the uniform estimates proved in the text.

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1 The OEIS problem, the source audit, and the scope of the results

1.1 One fundamental family, rather than unrelated sequences

Throughout, m and N are positive integers and $m \rightarrow \infty$. Define

$$F_m(q) = \prod_{j=1}^m \frac{1}{1-q^j}, \quad p_m(N) = [q^N]F_m(q), \quad V_m(N) = \frac{N^{m-1}}{m!(m-1)!}. \quad (1.1)$$

The equivalence between the two partition interpretations follows by conjugating Ferrers diagrams. This is a standard, widely useful family of restricted partitions, not an isolated artificial recurrence. The same analytic problem occurs when the size N grows polynomially, exponentially, or factorially with the allowed number m of parts.

The following are the specific OEIS targets inspected for this report. The descriptions and posted leading formulas are recorded in the respective entries [1, 2, 3, 4].

Entry or family	Count	Role in this report
A238016	$p_m(m^k)$, an array	A posted general conjecture; cubic threshold
A238608	$p_m(m^3)$	Five corrections and inverse expansion
A258302–A258305	$p_m(cm^3)$, $c = 2, 3, 4, 5$	Uniform critical-scale family
A238609–A238615	$p_m(m^k)$, $4 \leq k \leq 10$	All-order power-column expansions
A238010; A237998	$p_m(q^m)$; $q = 2$	Exponential sectors and their inversion
A238000	$p_m(m^m)$	Superexponential sectors and a Lambert chart
A258670	$p_m((2m)!)$	Factorial-scale consequence

The general condition in A238016 is expressed using the nonstandard notation $f(m) \geq O(m^4)$. Interpreting the intended assertion as growth at least a positive constant times m^4 , it asserts $p_m(f(m)) \sim V_m(f(m))$. The exact criterion in Theorem 2.3 is instead $f(m)/m^3 \rightarrow \infty$. It covers, for example, $f(m) = \lceil m^3 \log m \rceil$, not merely quartic growth.

1.2 What is classical, and what this report actually adds

A condition remaining conjectural on an OEIS page does not establish that it is an open problem in the research literature. Canfield’s account of Szekeres’ formula records the Erdős–Lehner estimate

$$p_m(N) = \frac{1}{m!} \binom{N-1}{m-1} \exp(O(m^3/N)), \quad m = O(N^{1/3}), \quad (1.2)$$

in Step 5 and explains its composition-counting origin [5, pp. 4, 9–10]. In particular, the sufficiency of $N/m^3 \rightarrow \infty$ is already a classical consequence. The critical leading constant for $N = cm^3$ is explicitly posted in A238608 [2].

The polynomial part of a restricted partition function, called its first Sylvester wave, is also classical. Dilcher and Vignat give explicit formulas using higher-order Bernoulli polynomials [6]. Sills and Zeilberger give an algorithmic quasipolynomial approach [7]; O’Sullivan studies waves and their asymptotics in a different joint-growth regime [8].

The contribution here is a complete, reusable treatment of the following more precise package: finite elementary inequalities proving necessity as well as sufficiency; a uniform all-order theorem

with all other arcs controlled; explicit critical coefficients through order five; an expansion with errors small enough to resolve q^{-m} sectors; explicit inverse charts; and a probabilistic boundary law with its first correction. The retrieved sources were not found to list these particular higher coefficients or this particular probability correction. That limited observation is not an exhaustive bibliographic novelty certification.

1.3 Relation to the ProveIt repository

The inspected repository file

```
Analysis/FabiusFunction/Lean/FabiusFunction/
PartitionBoundedParts.lean
```

provides a genuine formal starting point [12]. In namespace `Fabius`, it defines `boundedCount`, identifies restricted partitions with multiplicity vectors through `card_restricted_le_eq`, and proves the finite-product generating function in `hasSum_boundedCount_mul_pow`. The present analytic arguments can begin from precisely that generating function. We inspected the source but did not run the repository’s Lean build; none of the new asymptotic or probabilistic theorems is represented as already formalized.

1.4 Provenance and place in the collection

[write] **Added 1 October 2026, batch 74.** This report is a single manuscript, printed in full: manuscript 05 of batch 74 of ProveIt’s incoming-reports intake, delivered as the archive `OEIS_Restricted_Partitions_Cubic_Boundary.zip` in commit `c664fc2f0` and placed in the research-report collection, among the OEIS-sequence asymptotics reports, in commit `b669cff87`. It names no ProveIt commit: its repository input was a direct reading of the main-branch file `PartitionBoundedParts.lean`. At the placement commit that file still has the namespace `Fabius` (line 22) and the declarations `card_restricted_le_eq` (line 57), `boundedCount` (line 85) and `hasSum_boundedCount_mul_pow` (line 90), as described above. Those declarations are formal; nothing in this report is. Placement in the collection, beside that Lean development, confers no formal status on any statement here. The writing step changed only the labels (every label now carries the prefix `rpc`), added these [write] notes, three labels and two references [10, 11]; no statement or proof was altered.

The nearest report in the collection is `a097356-sqrt-restricted-partitions` [11], on the same counts $p_m(N)$ at the quadratic scale $N \asymp m^2$ (its parameter is $\alpha = N/m^2$, held in a compact set). This report works at $N \asymp m^3$ and beyond, which is the end $\alpha \rightarrow \infty$ of that report’s research question “Uniform limits as the saddle moves to zero or infinity”. It does not answer that question: no expansion here is matched uniformly to the quadratic regime, and the two reports share no proposition. The partition reports `a033552-catalan-partitions` (parts restricted to Catalan numbers) and `a022629-distinct-partition-norms` (the products $\prod(1 + k^a q^k)$) concern different products. Sufficiency in the cubic criterion is the classical Erdős–Lehner estimate, as §1 says; the leading constant of A238608 is posted in that entry.

2 An elementary sharp criterion

Write $R_m(N) = p_m(N)/V_m(N)$. The scale m^3 can be detected without saddle points, Bernoulli numbers, or complex analysis.

Lemma 2.1 (Lattice-volume bounds). *For $m \geq 1$ and $N \geq 1$, put $A_m = m(m+1)/2 - 1$. Then*

$$1 \leq R_m(N) \leq \left(1 + \frac{A_m}{N}\right)^{m-1}. \quad (2.1)$$

Proof. The multiplicity of the part 1 is determined by the other multiplicities. Consequently $p_m(N)$ is the number of integer points in

$$\mathcal{S}_N = \left\{ (x_2, \dots, x_m) \in \mathbb{R}_{\geq 0}^{m-1} : \sum_{j=2}^m jx_j \leq N \right\}.$$

This simplex has volume $N^{m-1}/((m-1)! \prod_{j=2}^m j) = V_m(N)$. Every point of $\mathcal{S}_N$ lies in the unit box based at its coordinatewise floor, a feasible integer point. Conversely, every unit box based at a feasible point lies in $\mathcal{S}_{N+A_m}$. The boxes have disjoint interiors and unit volume. Taking volumes proves both bounds. For $m = 1$ the statement is the zero-dimensional identity $p_1(N) = 1$. $\square$

Lemma 2.2 (A transposition lower bound). *For $m \geq 3$ and $N \geq 1$,*

$$R_m(N) \geq \prod_{r=1}^{m-1} \left(1 + \frac{r}{N}\right) + \frac{m(m-1)^2}{4N} \geq 1 + \frac{m(m-1)^2}{4N}. \quad (2.2)$$

Proof. Let the symmetric group act on weak compositions $(x_1, \dots, x_m)$ of N by permuting coordinates. Its orbits are partitions into at most m parts. Burnside's identity gives

$$p_m(N) = \frac{1}{m!} \sum_{\sigma \in S_m} |\text{Fix}(\sigma)|.$$

The identity fixes $\binom{N+m-1}{m-1}$ compositions, which contributes the product in (2.2) after normalization by $V_m(N)$.

A transposition requires two coordinates to be equal. Its fixed-point count is the number of solutions of

$$2y + z_1 + \dots + z_{m-2} = N, \quad y, z_i \geq 0.$$

Eliminate z_{m-2} and apply the lower unit-box argument to the resulting simplex in dimension $m-2$. The count is at least $N^{m-2}/(2(m-2)!)$. There are $\binom{m}{2}$ transpositions. Their total contribution, divided by $m!V_m(N)$, is $m(m-1)^2/(4N)$. All other terms in Burnside's sum are nonnegative. $\square$

Theorem 2.3 (Exact threshold). *For every positive-integer sequence $N = N(m)$,*

$$p_m(N) \sim V_m(N) \iff N/m^3 \rightarrow \infty. \quad (2.3)$$

Moreover, when $m^3/N \rightarrow 0$, $R_m(N) = 1 + O(m^3/N)$, whereas $N/m^3 \rightarrow 0$ implies $R_m(N) \rightarrow \infty$.

Proof. By (2.1), $0 \leq \log R_m(N) \leq (m-1)A_m/N = O(m^3/N)$. This proves sufficiency and its error estimate. If $R_m(N) \rightarrow 1$, (2.2) forces $m(m-1)^2/N \rightarrow 0$, which is equivalent to $m^3/N \rightarrow 0$. The same lower bound proves the last assertion. $\square$

A useful extension of the lower bound also explains the critical exponential. For $0 \leq r \leq \lfloor (m-1)/2 \rfloor$, retain the permutations consisting of r disjoint transpositions and $m-2r$ fixed points. The identical simplex argument gives

$$R_m(N) \geq \sum_{r=0}^{\lfloor (m-1)/2 \rfloor} \frac{m^{2r} (m-1)^r}{4^r r! N^r}. \quad (2.4)$$

Here $x^r = x(x-1)\cdots(x-r+1)$ and $x^0 = 1$. For $N \sim cm^3$, any fixed finite truncation tends to the corresponding truncation of $\exp(1/(4c))$. Thus

$$\liminf_{m \rightarrow \infty} R_m(N) \geq e^{1/(4c)}.$$

An upper bound with the same constant needs more information; it is supplied by the uniform expansion below. Formula (2.4) is not itself an assertion that all remaining permutation types can simply be discarded.

3 A uniform centered expansion to every fixed order

3.1 Centering and the coefficient recurrence

Set

$$S_r(m) = \sum_{j=1}^m j^r, \quad M = N + \frac{S_1(m)}{2} = N + \frac{m(m+1)}{4}. \quad (3.1)$$

The exact factorization

$$e^{Nt} F_m(e^{-t}) = \frac{e^{Mt}}{m! t^m} \mathcal{A}_m(t), \quad \mathcal{A}_m(t) = \prod_{j=1}^m \frac{jt}{2 \sinh(jt/2)} \quad (3.2)$$

is the reason for choosing this center. The remaining factor is even. Its logarithm, in a neighborhood of zero, is

$$\log \mathcal{A}_m(t) = \sum_{r \geq 1} \gamma_r(m) t^{2r}, \quad \gamma_r(m) = -\frac{B_{2r} S_{2r}(m)}{2r(2r)!}. \quad (3.3)$$

We use $B_2 = 1/6$, $B_4 = -1/30$, and $B_6 = 1/42$. Formula (3.3) follows by differentiating $\log(z/(2 \sinh(z/2)))$ and integrating its Bernoulli expansion at zero.

Define $a_j(m)$ by $\mathcal{A}_m(t) = \sum_{j \geq 0} a_j(m) t^{2j}$. Then

$$a_0 = 1, \quad j a_j = \sum_{r=1}^j r \gamma_r a_{j-r}. \quad (3.4)$$

The first coefficients are

$$a_1 = -\frac{S_2}{24}, \quad (3.5)$$

$$a_2 = \frac{S_2^2}{1152} + \frac{S_4}{2880}, \quad (3.6)$$

$$a_3 = -\frac{S_2^3}{82944} - \frac{S_2 S_4}{69120} - \frac{S_6}{181440}. \quad (3.7)$$

Each a_j is a rational polynomial in m of degree at most $3j$. Indeed S_{2r} has degree $2r+1 \leq 3r$, and the recurrence preserves the bound.

Theorem 3.1 (Uniform centered all-order expansion). *Fix $c_0 > 0$ and an integer $K \geq 0$. Uniformly over integer pairs with $N \geq c_0 m^3$, as $m \rightarrow \infty$,*

$$p_m(N) = \frac{M^{m-1}}{m!(m-1)!} \left\{ \sum_{j=0}^K \frac{a_j(m)(m-1)^{2j}}{M^{2j}} + O_{K,c_0}(m^{-K-1}) \right\}. \quad (3.8)$$

In particular, the j th displayed term is $O_{j,c_0}(m^{-j})$. The theorem concerns fixed truncation orders; it does not assert convergence of the resulting infinite asymptotic series.

3.2 The minor-arc estimate, including very large N

We give the details because a formal expansion at $q = 1$ alone does not justify (3.8). Other roots of unity must be controlled uniformly while the number of factors grows.

Lemma 3.2 (Decay on a Cauchy circle). *Let $0 < \tau \leq C_0 m^{-2}$, and put*

$$\rho_m(\theta) = \frac{|F_m(e^{-\tau-i\theta})|}{F_m(e^{-\tau})}.$$

For all sufficiently large m there are positive constants b, δ, C , with dependence on C_0 allowed, such that

$$\rho_m(\theta) \leq \left(1 + b(\theta/\tau)^2\right)^{-m/2}, \quad |\theta| \leq 1/m, \quad (3.9)$$

$$\rho_m(\theta) \leq (Cm\tau)^{\delta m}, \quad 1/m \leq |\theta| \leq \pi. \quad (3.10)$$

Proof. For each factor,

$$\frac{1 - e^{-j\tau}}{|1 - e^{-j\tau - ij\theta}|} = \left(1 + \frac{4e^{-j\tau} \sin^2(j\theta/2)}{(1 - e^{-j\tau})^2}\right)^{-1/2}.$$

If $|\theta| \leq 1/m$, then $|j\theta| \leq 1$ and $j\tau = O(1/m)$. The inequalities $|\sin(j\theta/2)| \geq cj|\theta|$ and $1 - e^{-j\tau} \leq j\tau$ prove (3.9) after multiplication.

For (3.10), a fixed positive fraction of the indices $j \leq m$ have $|\sin(j\theta/2)|$ bounded away from zero, uniformly in this interval. Here is an explicit justification. For $1/m \leq |\theta| \leq 4\pi/m$, choose j between $m/(16\pi)$ and $m/(8\pi)$; the arguments stay in a compact subinterval of $(0, \pi)$. For $4\pi/m \leq |\theta| \leq \pi$, the geometric-sum bound gives

$$\sum_{j=1}^m \sin^2(j\theta/2) \geq \frac{m}{2} - \frac{1}{2|\sin(\theta/2)|} \geq \frac{3m}{8}.$$

This forces a positive fraction of the summands to exceed a fixed positive constant. Each corresponding factor is at most $Cm\tau$; every other factor is at most one. Taking the minimum of the constants from the two cases proves the assertion. $\square$

3.3 Proof of the uniform theorem

Proof of Theorem 3.1. Cauchy's formula, with $q = e^{-t}$, is

$$p_m(N) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{N(\tau+i\theta)} F_m(e^{-\tau-i\theta}) d\theta. \quad (3.11)$$

Choose $\tau = m/M$ and the central interval $|\theta| \leq \tau m^{-2/5}$. We use three estimates.

Analytic Taylor remainder. For a sufficiently small absolute $\varepsilon > 0$, $\mathcal{A}_m(t)$ is analytic and uniformly bounded on $|t| \leq \varepsilon m^{-3/2}$. In fact $j|t| = O(m^{-1/2})$ there, and the sum of the absolute logarithmic bounds is $O(S_2(m)|t|^2) = O(1)$. Cauchy's coefficient estimate and evenness imply

$$\mathcal{A}_m(t) = \sum_{j=0}^K a_j(m) t^{2j} + O_K\left(m^{3K+3} |t|^{2K+2}\right)$$

inside half that disk. The central interval is contained in this smaller disk for all sufficiently large m , uniformly in $N \geq c_0 m^3$.

Integrating the central remainder. For fixed nonnegative r and $m > r + 2$,

$$\int_{-\infty}^{\infty} |\tau + i\theta|^{-m+r} d\theta = \tau^{1-m+r} \sqrt{\pi} \frac{\Gamma((m-r-1)/2)}{\Gamma((m-r)/2)}.$$

The gamma quotient is $O_r(m^{-1/2})$. Together with Stirling's formula, this shows that integrating the Taylor remainder in (3.2), relative to $M^{m-1}/(m!(m-1)!)$, costs at most

$$O_K(m^{3K+3}\tau^{2K+2}) = O_{K,c_0}(m^{-K-1}).$$

The elementary inverse-Laplace identity

$$\frac{1}{2\pi i} \int_{\tau-i\infty}^{\tau+i\infty} e^{Mt} t^{-d} dt = \frac{M^{d-1}}{(d-1)!}, \quad d \geq 2,$$

follows by closing the contour to the left and taking the residue at zero. Apply it with $d = m - 2j$. Extending each central polynomial integral to this line changes it by an exponentially small quantity: the gamma kernel outside $|\theta| \leq \tau m^{-2/5}$ has a tail bounded by $\exp(-b_1 m^{1/5})$ times its natural integral size. This gives exactly the falling factorial in (3.8).

Discarding the rest of the original circle. On $\tau m^{-2/5} \leq |\theta| \leq 1/m$, (3.9) and the change of variable $v = \theta/\tau$ give a relative contribution $O(\exp(-b_2 m^{1/5}))$. On the remaining arcs, the relative contribution is bounded by

$$C \frac{\sqrt{m}}{\tau} (Cm\tau)^{\delta m}. \quad (3.12)$$

The factor $\sqrt{m}/\tau$ comes from normalizing the real-axis integrand by its gamma-kernel integral. It cannot be ignored when N is extremely large. For $\delta m > 1$, the right side is increasing in τ . Substituting $\tau \leq C_0/m^2$ bounds it by $m^{5/2}(C_1/m)^{\delta m}$, which is smaller than every fixed inverse power of m . Thus the argument is uniform even for factorial or faster growth of N , rather than merely for N bounded by a power of m .

Combining the three estimates proves (3.8). $\square$

Remark 3.3 (What this theorem does not assert). For fixed m , the polynomial part of $p_m(N)$ is its first Sylvester wave. The coefficient algebra above is consistent with that classical polynomial. The theorem gives uniform finite-order asymptotics of the entire count, not just of a polynomial formally extracted at one singularity. It does not identify each exponentially small root-of-unity wave, provide a Stokes analysis, or specify an optimal truncation order growing with m .

4 Critical constants and explicit OEIS expansions

Corollary 4.1 (Critical crossover). *If $N/m^3 \rightarrow c \in (0, \infty)$, then*

$$\frac{p_m(N)}{V_m(N)} \rightarrow e^{1/(4c)}. \quad (4.1)$$

Proof. The $K = 0$ case of Theorem 3.1 is $p_m(N) = M^{m-1}/(m!(m-1)!(1 + O(1/m)))$. Furthermore,

$$(m-1) \log(M/N) = (m-1) \log \left(1 + \frac{m(m+1)}{4N} \right) \rightarrow \frac{1}{4c}.$$

$\square$

4.1 A general all-order logarithmic expansion

Write $c = N/m^3$; this notation is permitted even when c varies with m . For every fixed K , uniformly for $c \geq c_0 > 0$,

$$\begin{aligned} \log p_m(cm^3) &= (m-3) \log m + 2m + (m-1) \log c - \log(2\pi) + \frac{1}{4c} \\ &\quad + \sum_{r=1}^K \frac{L_r(c)}{m^r} + O_{K,c_0}(m^{-K-1}). \end{aligned} \quad (4.2)$$

Here and below the left side is used only when cm^3 is an integer. The $L_r(c)$ are rational polynomials in $1/c$. Equivalently,

$$p_m(cm^3) = \frac{e^{2m+1/(4c)} c^{m-1} m^{m-3}}{2\pi} \left(\sum_{r=0}^K \frac{b_r(c)}{m^r} + O_{K,c_0}(m^{-K-1}) \right), \quad (4.3)$$

where

$$b_0 = 1, \quad kb_k = \sum_{r=1}^k r L_r b_{k-r}. \quad (4.4)$$

To obtain (4.2), take the logarithm of (3.8), substitute $M = cm^3(1 + (m^{-1} + m^{-2})/(4c))$, and use the classical Stirling expansion [9]:

$$\log \Gamma(m+1) + \log \Gamma(m) \sim 2m \log m - 2m + \log(2\pi) + 2 \sum_{r \geq 1} \frac{B_{2r}}{2r(2r-1)m^{2r-1}}. \quad (4.5)$$

Every operation is justified to a fixed order by Theorem 3.1. Thus the coefficient algorithm is backed by an asymptotic remainder theorem, not only by formal manipulation.

The first five logarithmic coefficients are

$$L_1(c) = -\frac{48c^2 + 13}{288c^2}, \quad (4.6)$$

$$L_2(c) = -\frac{144c^2 + 6c - 7}{576c^3}, \quad (4.7)$$

$$L_3(c) = \frac{11520c^4 + 122400c^2 + 14400c - 8081}{2073600c^4}, \quad (4.8)$$

$$L_4(c) = \frac{21600c^3 - 50400c^2 - 7855c + 2861}{2073600c^5}, \quad (4.9)$$

$$\begin{aligned} L_5(c) &= -\frac{1}{21946982400c^6} (34836480c^6 + 304819200c^4 + 304819200c^3 \\ &\quad - 235282320c^2 - 42928704c + 11425901). \end{aligned} \quad (4.10)$$

As a short independent check of the first coefficient, the centered shift contributes $-1/(32c^2m)$, the a_1 term contributes $-1/(72c^2m)$, and Stirling contributes $-1/(6m)$. Their sum is (4.6).

4.2 Five corrections for A238608

At $c = 1$, the result becomes

$$p_m(m^3) = \frac{e^{2m+1/4}m^{m-3}}{2\pi} \left[1 - \frac{61}{288m} - \frac{37463}{165888m^2} + \frac{425075267}{3583180800m^3} \right. \\ \left. - \frac{21833043287}{4127824281600m^4} - \frac{2048618007321713}{58251856261939200m^5} + O(m^{-6}) \right]. \quad (4.11)$$

The leading factor is the one already posted in A238608 [2]. The additional terms and their remainder are the refinement. The logarithmic form is often more stable numerically:

$$\log \left(\frac{2\pi p_m(m^3)}{e^{2m+1/4}m^{m-3}} \right) = -\frac{61}{288m} - \frac{143}{576m^2} + \frac{140239}{2073600m^3} \\ - \frac{16897}{1036800m^4} - \frac{377689757}{21946982400m^5} + O(m^{-6}). \quad (4.12)$$

Substituting $c = 2, 3, 4, 5$ in the general formulas immediately gives the same order of refinement for A258302–A258305, without separate saddle analyses.

4.3 Higher power columns and an important rounding issue

For any fixed integer $k \geq 4$, substitute $c = m^{k-3}$ in (4.2). One obtains the complete inverse-power expansion of

$$p_m(m^k) \sim \frac{e^{2m}m^{(k-2)m-k}}{2\pi}, \quad (4.13)$$

the leading expression posted in the A238016 family [1]. For example, the quartic column A238609 satisfies

$$\log p_m(m^4) = (2m - 4) \log m + 2m - \log(2\pi) + \frac{1}{12m} - \frac{139}{480m^3} \\ - \frac{1}{96m^4} + \frac{1403}{20160m^5} + O(m^{-6}). \quad (4.14)$$

In particular, its first correction has the opposite sign from the cubic column's first correction.

If the intended count is $p_m(\lfloor cm^3 \rfloor)$ for a fixed nonintegral c , use $c_m = \lfloor cm^3 \rfloor / m^3$ in (4.2). Replacing c_m indiscriminately by c changes the logarithm by $O(m^{-2})$, which is large enough to corrupt coefficients beginning at the second correction. This is a discrete rounding effect, not a failure of the theorem.

5 Dilute expansions and genuine exponential sectors

The centered theorem is uniform in very large N , but its remainder $O(m^{-K-1})$ is not small enough to detect a term such as 2^{-m} . A separate expansion in $1/N$, with a remainder measured in m^3/N , is needed.

5.1 The uncentered coefficient system

Define

$$\mathcal{H}_m(t) = \prod_{j=1}^m \frac{jt}{1 - e^{-jt}} = \exp \left(\frac{S_1 t}{2} + \sum_{r \geq 1} \gamma_r t^{2r} \right) = \sum_{j \geq 0} h_j(m) t^j. \quad (5.1)$$

Put

$$\beta_j(m) = (m-1)^j h_j(m), \quad \beta_0 = 1. \quad (5.2)$$

If $\lambda_1 = S_1/2$, $\lambda_{2r} = \gamma_r$, and $\lambda_{2r+1} = 0$ for $r \geq 1$, then

$$jh_j = \sum_{r=1}^j r \lambda_r h_{j-r}. \quad (5.3)$$

Thus h_j and β_j have rational coefficients and degrees at most $2j$ and $3j$, respectively.

Theorem 5.1 (Uniform dilute expansion). *There is an absolute $\eta_0 > 0$ such that, for every fixed $K \geq 0$, uniformly for $0 < \eta = m^3/N \leq \eta_0$ and all sufficiently large m ,*

$$\frac{p_m(N)}{V_m(N)} = \sum_{j=0}^K \frac{\beta_j(m)}{N^j} + O_K((m^3/N)^{K+1}). \quad (5.4)$$

In particular, this holds along every sequence for which $m^3/N \rightarrow 0$.

Proof. Use (3.11) with $\tau = m/N = \eta/m^2$. The exact integrand now is $e^{Nt} \mathcal{H}_m(t)/(m!t^m)$. For a sufficiently small fixed $r > 0$, $\mathcal{H}_m$ is uniformly bounded on $|t| \leq r/m^2$: the linear logarithmic term is bounded, and all other terms have total absolute size $O(m^3|t|^2)$. Cauchy's estimate gives

$$\mathcal{H}_m(t) = \sum_{j=0}^K h_j t^j + O_K((m^2|t|)^{K+1})$$

on the disk of half this radius. Choose η_0 sufficiently small that τ lies well inside the disk and integrate over $|\theta| \leq r/(4m^2)$. The same gamma-kernel calculation as before bounds the relative remainder by $(m^2\tau)^{K+1} = \eta^{K+1}$. Each polynomial term gives $h_j(m-1)^j/N^j$ by inverse Laplace integration.

For completeness, the omitted arcs are small on this stronger scale, not merely small in $1/m$. On $r/(4m^2) \leq |\theta| \leq 1/m$, Lemma 3.2, after $v = \theta/\tau$, bounds the normalized tails by a polynomial factor in m times $(C\eta)^{m-1}$. For sufficiently small η_0 and m sufficiently large relative to K , this is $O_K(\eta^{K+1})$. On $1/m \leq |\theta| \leq \pi$, the analogous bound is

$$C \frac{m^{5/2}}{\eta} \left(\frac{C\eta}{m} \right)^{\delta m}.$$

After division by η^{K+1} , the positive remaining exponent of η allows substitution of η_0 , leaving a quantity tending rapidly to zero. These bounds also justify extending the polynomial integrals to the entire vertical line. This proves (5.4). $\square$

Taking the formal logarithm defines rational polynomials $D_j(m)$ through

$$D_j = \beta_j - \frac{1}{j} \sum_{r=1}^{j-1} r D_r \beta_{j-r}. \quad (5.5)$$

Theorem 5.1 justifies the corresponding logarithmic expansion with the same order of remainder. Its first terms are

$$D_1(m) = \frac{m(m^2 - 1)}{4}, \quad (5.6)$$

$$D_2(m) = -\frac{m(m^2 - 1)(13m^2 + 3m - 4)}{288}. \quad (5.7)$$

For example $h_2 = S_1^2/8 - S_2/24$, so $D_2 = (m - 1)(m - 2)h_2 - D_1^2/2$, an immediate exact check.

5.2 Exponential and superexponential specializations

For a fixed integer $q \geq 2$, put $a_q(m) = p_m(q^m)$ and $b = \log q$. The full fixed-order expansion is

$$\begin{aligned} \log a_q(m) &= bm(m - 1) - \log \Gamma(m + 1) - \log \Gamma(m) \\ &\quad + \sum_{j=1}^K D_j(m) e^{-jbm} + O_K(m^{3K+3} e^{-(K+1)bm}). \end{aligned} \quad (5.8)$$

This is an expansion in genuine exponentially separated sectors, rather than only a power series in $1/m$ renamed a transseries. The exact gamma background is retained so that its algebraic truncation error does not obscure the exponential corrections. In particular,

$$\begin{aligned} \log a_q(m) &= bm(m - 1) - \log \Gamma(m + 1) - \log \Gamma(m) \\ &\quad + \frac{m(m^2 - 1)}{4} q^{-m} - \frac{m(m^2 - 1)(13m^2 + 3m - 4)}{288} q^{-2m} + O(m^9 q^{-3m}). \end{aligned} \quad (5.9)$$

The leading term corresponds to A238010 and its columns, including A237998 when $q = 2$ [3].

For A238000, put $N = m^m$ in Theorem 5.1. The sectors are $m^{-jm} = \exp(-jm \log m)$, with the same polynomial coefficients $D_j(m)$. For A258670, put $N = (2m)!$; its sectors are $((2m)!)^{-j}$ [4]. In particular,

$$p_m((2m)!) = \frac{((2m)!)^{m-1}}{m!(m-1)!} \left(1 + \frac{m(m^2 - 1)}{4(2m)!} + O\left(\frac{m^6}{((2m)!)^2}\right) \right).$$

There is no need to solve a new saddle equation for each such sequence. The full collection follows from one uniform dilute theorem.

6 Inverting the asymptotic expansions

6.1 What inversion means for an integer sequence

A sequence has no canonical smooth interpolation. We therefore use an *asymptotic inverse chart*: a smooth expression $I(y)$ that, at the exact sample values $y = a(m)$, recovers m with a specified asymptotic error. The same expression can approximate an integer threshold inverse $Q(y) = \min\{m : a(m) \geq y\}$, but generally only up to an $O(1)$ rounding ambiguity. An exponentially accurate sample-point chart does not magically supply an exponentially accurate integer-valued function for all real y .

Each fixed truncation of a forward logarithmic profile is smooth for large real x and can be inverted there. Raising the truncation order produces consistent inverse coefficients. At sample

values the forward remainder, divided by the derivative of the leading profile, controls the inverse error. This is the precise sense of the all-order inversions used below.

[write] **Added 1 October 2026, batch 74: an instance of repository results.** The distinction drawn here between a sample-point chart and the integer threshold $Q(y)$, with its $O(1)$ rounding ambiguity, is the content of the staircase theorem `p0:thm:staircase` of the transseries-and-inversion volume [10]: $Q = \lceil \nu \rceil$ for any increasing interpolation ν , and a chart with error at most η determines Q exactly when the separation condition of its part (2) holds. The inversions of this section are instances of that volume's inversion apparatus, and no novelty is claimed for the method.

6.2 The cubic family: Lambert W and two explicit corrections

Fix $c > 0$, and consider $a_c(m) = p_m(cm^3)$ on admissible integer arguments. Write

$$b_c = 2 + \log c, \quad D_c = \frac{1}{4c} - \log(2\pi c).$$

Equation (4.2) has the form

$$\log a_c(m) = m \log m + b_c m - 3 \log m + D_c + \frac{L_1(c)}{m} + \frac{L_2(c)}{m^2} + \dots \quad (6.1)$$

For large y , set $L = \log y$ and

$$w = W(ce^2 L), \quad x_0 = \frac{L}{w}. \quad (6.2)$$

Here W is the positive real solution of $we^w = z$. Then $\log x_0 + b_c = w$, and consequently $x_0 \log x_0 + b_c x_0 = L$ exactly. Define

$$d_0 = \frac{3 \log x_0 - D_c}{w + 1}, \quad d_1 = \frac{3d_0 - d_0^2/2 - L_1(c)}{w + 1}. \quad (6.3)$$

Theorem 6.1 (Cubic inverse chart). *At $y = a_c(m)$, with c fixed,*

$$m = x_0 + d_0 + \frac{d_1}{x_0} + O_c(x_0^{-2}). \quad (6.4)$$

There is an all-order expansion $x_0 + \sum_{r \geq 0} d_r(w; c)x_0^{-r}$, in the fixed-order asymptotic sense. Its coefficients are obtained recursively by Taylor substitution in (6.1); they are rational functions of w with constants depending on c .

Proof. Let $f_0(x) = x \log x + b_c x$. At x_0 , one has $f'_0 = w + 1$ and $f''_0 = 1/x_0$. Substitute $x = x_0 + d_0 + d_1/x_0$ in (6.1). The constant residual is $(w + 1)d_0 - 3 \log x_0 + D_c$, and the coefficient of x_0^{-1} is $(w + 1)d_1 + d_0^2/2 - 3d_0 + L_1(c)$. Both vanish by (6.3). The remaining residual is $O_c(x_0^{-2})$, whereas the derivative is asymptotic to $w + 1$. The mean-value theorem proves the stated, slightly weaker, $O_c(x_0^{-2})$ inverse error.

At any further fixed order, the new unknown coefficient appears multiplied by $w + 1$; everything else is already known. Division by $w + 1$ completes an induction. Since $\log x_0 = w - b_c$, the recursion stays within rational functions of w . The forward expansion can be taken to arbitrarily high fixed order by Theorem 3.1. $\square$

For A238608 specifically,

$$d_0 = 3 - \frac{37/4 - \log(2\pi)}{w+1}, \quad d_1 = \frac{3d_0 - d_0^2/2 + 61/288}{w+1}, \quad w = W(e^2 \log y).$$

These corrections are important even for moderately large indices: the raw Lambert term can miss the index by nearly two while the corrected expression is already accurate to about 10^{-4} at $m = 60$.

[write] **Added 1 October 2026, batch 74: an instance of repository results.** The core $x \log x + b_c x = L$, solved by the principal branch as in (6.2), is the Lambert core of the inverse gamma function in **p6:thm:gamma** of the transseries-and-inversion volume [10] (there $T \log T - T = R$ with $T = R/W_0(R/e)$), shifted by the constant b_c ; the corrections $d_0, d_1, \dots$ are its recursive perturbed inversion **p0:thm:perturbed-inversion**. The exponential columns below use the same perturbed-inversion step with the small parameter e^{-bX} . Theorem 6.1 and (6.8) are instances of that apparatus.

6.3 Exponential columns: algebraic and invisible corrections

For $q \geq 2$ fixed, define the exact smooth background

$$G_q(x) = bx(x-1) - \log \Gamma(x+1) - \log \Gamma(x), \quad b = \log q. \quad (6.5)$$

Stirling's expansion yields

$$G_q(x) = bx^2 - 2x \log x + (2-b)x - \log(2\pi) - \frac{1}{6x} + \frac{1}{180x^3} - \dots$$

Put

$$s = \sqrt{L/b}, \quad d = \frac{\log s}{b} + \frac{b-2}{2b}, \quad e = \frac{d^2}{2} + \frac{d}{b} + \frac{\log(2\pi)}{2b}. \quad (6.6)$$

Taylor substitution gives the algebraic inverse chart

$$m = s + d + \frac{e}{s} + O_q\left(\frac{(\log s)^3}{s^2}\right), \quad y = a_q(m). \quad (6.7)$$

For example, the coefficient of s cancels because $2bd = 2 \log s + b - 2$, and the constant residual cancels because $2be = bd^2 + 2d + \log(2\pi)$.

To expose an exponentially small inverse correction, let X be the exact large solution of $G_q(X) = \log y$. This background inverse is well defined for large y since

$$G'_q(X) = b(2X-1) - \psi(X+1) - \psi(X) \sim 2bX > 0.$$

Equation (5.8) and one implicit Taylor step give

$$m = X - \frac{D_1(X)e^{-bX}}{G'_q(X)} + O_q(X^5 e^{-2bX}), \quad y = a_q(m). \quad (6.8)$$

The sign is negative: the positive count correction raises $\log a_q$ above the gamma background, so the corrected index at fixed y is smaller. The $O(m^6 e^{-2bm})$ forward error divided by $G'_q(m) \asymp m$ gives the displayed safe inverse error; the Taylor cross terms are smaller.

All further exponential sectors can be computed recursively. For example, write $m = X + u_1(X)e^{-bX} + u_2(X)e^{-2bX} + \dots$. Then

$$u_1 = -D_1/G'_q, \quad (6.9)$$

$$u_2 = -\frac{D_2 + (D'_1 - bD_1)u_1 + G''_q u_1^2/2}{G'_q}. \quad (6.10)$$

Theorem 5.1 supplies the required forward remainder at every fixed sector order. Retaining only finitely many terms of Stirling in G_q would not justify (6.8): that algebraic error would swamp the quantity being resolved.

6.4 The diagonal sequence A238000

The same procedure gives a particularly simple leading inverse for A238000. Here

$$\log p_m(m^m) = (m^2 - 3m) \log m + 2m - \log(2\pi) + O(1/m),$$

with exponentially smaller corrections in m^{-m} supplied by Section 5. If

$$x_0 = \sqrt{\frac{2L}{W(2L)}},$$

then $x_0^2 \log x_0 = L$, and a first correction is

$$m = x_0 + \frac{3 \log x_0 - 2}{2 \log x_0 + 1} + O(x_0^{-1}) \quad \text{at } y = p_m(m^m). \quad (6.11)$$

This follows by dividing the leading residual $-3x_0 \log x_0 + 2x_0$ by the derivative $x_0(2 \log x_0 + 1)$. It provides another inverse chart from the same partition theorem, rather than an unrelated inversion example.

7 A Poisson boundary law in the conjugate partition model

7.1 Which repeated parts are being counted

Choose a partition of N into *at most* m parts uniformly, and pad it with zeros to obtain

$$0 \leq x_1 \leq \dots \leq x_m, \quad x_1 + \dots + x_m = N.$$

This is the conjugate of the original bounded-part-size model. The distinction matters: partitions whose part sizes are bounded by m typically have many repeated small parts, and the theorem below is not a claim about those multiplicities.

Set $d_1 = x_1$ and $d_i = x_i - x_{i-1}$ for $i \geq 2$. Then

$$\sum_{i=1}^m (m - i + 1) d_i = N.$$

Let

$$Z_m = \#\{i : 2 \leq i \leq m, d_i = 0\},$$

the number of equal adjacent pairs in the padded partition. A gap equal to zero corresponds to a zero multiplicity for one of the weights $1, \dots, m-1$ in the generating product. Its probability generating function is exactly

$$\mathbb{E}[u^{Z_m}] = \frac{[q^N]F_m(q) \prod_{j=1}^{m-1} (u + (1-u)q^j)}{p_m(N)}. \quad (7.1)$$

Indeed for each marked weight the sum over a gap is $u + q^j + q^{2j} + \dots = (u + (1-u)q^j)/(1-q^j)$.

7.2 A deletion estimate and the limit distribution

Lemma 7.1 (Deleting a fixed number of factors). *Fix k . Uniformly over k -element sets $S \subseteq \{1, \dots, m\}$ and N/m^3 in a compact subinterval of $(0, \infty)$,*

$$\frac{[q^N]F_m(q) \prod_{j \in S} (1 - q^j)}{p_m(N)} = \left(\prod_{j \in S} j \right) \frac{(m-1)^k}{M^k} (1 + O_k(m^{-1})). \quad (7.2)$$

Proof. On the central interval of Section 3,

$$\prod_{j \in S} (1 - e^{-jt}) = \left(\prod_{j \in S} j \right) t^k (1 + O_k(m|t|)) = \left(\prod_{j \in S} j \right) t^k (1 + O_k(m^{-1})).$$

Inserting this factor in the contour proof changes t^{-m} to t^{-m+k} , so the inverse-Laplace integral contributes $(m-1)^k/M^k$. Both $\mathcal{A}_m(t)$ and the denominator count have relative error $O(1/m)$ at this precision. On the other arcs the new product has modulus at most 2^k . Since k is fixed and $\tau \asymp m^{-2}$, normalizing by the smallest possible main deleted-factor contribution loses only a fixed power of m , harmless compared with the arc decay. All bounds are uniform in the choice of S . $\square$

Theorem 7.2 (Poisson repeated-part boundary). *If $N/m^3 \rightarrow c \in (0, \infty)$, then*

$$Z_m \xrightarrow{d} \text{Poisson}\left(\frac{1}{2c}\right). \quad (7.3)$$

With probability tending to one, the partition has exactly m positive parts and no part occurs three or more times. Thus Z_m also agrees, with probability tending to one, with the number of distinct positive part sizes that occur twice.

Proof. For every fixed k , the factorial-moment identity from (7.1) and Lemma 7.1 give

$$\mathbb{E}[Z_m^k] = k! \frac{(m-1)^k}{M^k} \sum_{\substack{S \subseteq \{1, \dots, m-1\} \\ |S|=k}} \prod_{j \in S} j (1 + O_k(m^{-1})).$$

For fixed k , the elementary symmetric sum here equals

$$\frac{1}{k!} \left(\sum_{j=1}^{m-1} j \right)^k (1 + O_k(m^{-1})).$$

To see this, expand the power and remove tuples with a repeated index; their relative total is bounded by a constant times $\sum j^2/(\sum j)^2 = O(1/m)$. Therefore

$$\mathbb{E}[Z_m^k] \longrightarrow (1/(2c))^k.$$

These are the factorial moments of the asserted Poisson law. For clarity, the first moment gives tightness; the next higher moments give uniform integrability of every fixed power along subsequences. Any subsequential limit therefore has the Poisson moments. The Poisson distribution is moment-determinate, as its moment generating function is finite in a neighborhood of zero. This proves (7.3).

The event $d_1 = 0$ is a single deletion of weight m , and hence has probability $m(m-1)/M(1+O(1/m)) = O(1/m)$. A triple repetition requires two consecutive gaps to vanish. By (7.2), summing these two-deletion probabilities over at most m adjacent pairs costs

$$O\left(\frac{m^2}{M^2} \sum_{j=1}^{m-1} j(j+1)\right) = O(m^{-1}).$$

Outside these two exceptional events there are m positive parts and only double repetitions, proving the last assertion. $\square$

7.3 An explicit correction to the Poisson generating function

The limit can be refined without assuming independence of the gaps.

Theorem 7.3 (First finite-size probability correction). *Let $c = N/m^3$ range over a compact subinterval of $(0, \infty)$. Uniformly for real $0 \leq u \leq 1$,*

$$\mathbb{E}[u^{Z_m}] = e^{(u-1)/(2c)} \left[1 - \frac{1}{m} \left(\frac{u-1}{c} + \frac{7u(u-1)}{24c^2} \right) + O_c(m^{-2}) \right]. \quad (7.4)$$

The constant in the error can be chosen uniformly on the indicated compact c interval. In particular,

$$\mathbb{P}(Z_m = 0) = e^{-1/(2c)} \left(1 + \frac{1}{cm} + O_c(m^{-2}) \right). \quad (7.5)$$

Proof. Put $v = u - 1$, $S'_r = \sum_{j=1}^{m-1} j^r$, and $M_v = M + vS'_1$. Near zero,

$$\log \prod_{j=1}^{m-1} (u + (1-u)e^{-jt}) = vS'_1 t - \frac{v+v^2}{2} S'_2 t^2 + O(m^4 |t|^3),$$

uniformly for $0 \leq u \leq 1$. After combining this with (3.3), define

$$C_v = \frac{S_2}{24} + \frac{v+v^2}{2} S'_2.$$

The same centered contour proof, now at $\tau = m/M_v$, yields for the numerator of (7.1)

$$\frac{M_v^{m-1}}{m!(m-1)!} \left(1 - C_v \frac{(m-1)(m-2)}{M_v^2} + O(m^{-2}) \right).$$

The error follows from $m^4|t|^3 + m^6|t|^4 = O(m^{-2})$ on the central scale. The old arc estimates remain valid: for $0 \leq u \leq 1$ and $|q| \leq 1$, every multiplier $u + (1-u)q^j$ has modulus at most one. Also $M_v = M + O(m^2) > 0$ and the change of real-axis normalization costs only a bounded factor.

Divide by the same formula at $v = 0$ and take a logarithm. With $N = cm^3$,

$$(m-1)\log(M_v/M) = \frac{v}{2c} - \frac{1}{m} \left(\frac{v}{c} + \frac{v+v^2}{8c^2} \right) + O(m^{-2}),$$

$$-C_v \frac{(m-1)(m-2)}{M_v^2} + C_0 \frac{(m-1)(m-2)}{M^2} = -\frac{v+v^2}{6c^2m} + O(m^{-2}).$$

Since $1/8 + 1/6 = 7/24$, exponentiation proves (7.4). Setting $u = 0$ proves (7.5). $\square$

There is a useful independent exact identity at $u = 0$. Subtract the staircase $(0, 1, \dots, m-1)$ from a strictly increasing nonnegative partition. The result is an arbitrary weakly increasing nonnegative partition of $N - T$, where $T = m(m-1)/2$. Thus

$$\mathbb{P}(Z_m = 0) = \frac{p_m(N - T)}{p_m(N)}. \quad (7.6)$$

Inserting the centered expansion into this ratio checks both the limit $e^{-1/(2c)}$ and the first correction $1/(cm)$. Higher pgf corrections can be generated by retaining further terms in $\log(u + (1-u)e^{-z})$ and using the same coefficient recursions. We do not claim a total-variation rate or a uniform complex- u expansion from the real interval estimate alone.

8 Exact computation and reproducible diagnostics

8.1 Independent coefficient extraction

The package's `verify.py` computes integer coefficients by the standard product update

$$P_0(0) = 1, \quad P_0(n) = 0 \ (n > 0), \quad P_j(n) = P_{j-1}(n) + P_j(n-j),$$

where negative arguments contribute zero. A second implementation uses the logarithmic derivative of (1.1):

$$np_m(n) = \sum_{r=1}^n \left(\sum_{\substack{d|r \\ d \leq m}} d \right) p_m(n-r). \quad (8.1)$$

These two methods agree for $1 \leq m \leq 12$ and $0 \leq n \leq 200$. The finite inequalities in Section 2 are tested after clearing denominators, so their verification uses exact integer arithmetic. All displayed initial A238608 values through $m = 16$ agree with the inspected OEIS entry [2].

The complete run used 90-decimal-digit floating-point arithmetic only after the exact counts were produced. It performed 9,230 exact assertions and recorded 258 count samples, with $m \leq 60$ and coefficient arrays extending to $N = 320000$. The sample families include cm^3 for $1 \leq c \leq 5$, m^4 , and 2^m . These are finite tests; the all-order and uniform statements rest on the contour estimates, not on extrapolating the tables.

8.2 Accuracy of the cubic expansion

Let $A_K(m)$ be the right side of (4.11), truncated after m^{-K} . The following entries are signed relative errors $A_K(m)/p_m(m^3) - 1$ from the included exact computation.

m	$K = 0$	$K = 1$	$K = 3$	$K = 5$
10	$2.38780 \cdot 10^{-2}$	$2.19166 \cdot 10^{-3}$	$8.72320 \cdot 10^{-7}$	$-2.93140 \cdot 10^{-8}$
20	$1.12656 \cdot 10^{-2}$	$5.55991 \cdot 10^{-4}$	$4.40927 \cdot 10^{-8}$	$-4.51323 \cdot 10^{-10}$
40	$5.46413 \cdot 10^{-3}$	$1.40056 \cdot 10^{-4}$	$2.41567 \cdot 10^{-9}$	$-7.04814 \cdot 10^{-12}$
60	$3.60523 \cdot 10^{-3}$	$6.24068 \cdot 10^{-5}$	$4.54363 \cdot 10^{-10}$	$-6.18667 \cdot 10^{-13}$

The full CSV includes $K = 2$, the normalized exact count, and the other values of c . A small error at one index alone would not prove an expansion; these data are useful as stringent checks on coefficient signs, normalization, and omitted shifts.

8.3 Probability and inverse diagnostics

For $c = 1$, the limiting mean is $1/2$, the limiting zero probability is $e^{-1/2} \approx 0.60653066$, and the limiting pgf at $u = 1/2$ is $e^{-1/4} \approx 0.77880078$. Exact finite values are

m	$\mathbb{E}Z_m$	$\mathbb{P}(Z_m = 0)$	$\mathbb{E}[2^{-Z_m}]$
10	0.3846890512	0.6682085339	0.8213708910
20	0.4390680097	0.6371577383	0.8006057040
40	0.4686467899	0.6217743594	0.7898304769
60	0.4788932680	0.6166760199	0.7861818675

The first-corrected pgf at $u = 1/2$ has relative error about $7.04 \cdot 10^{-5}$ at $m = 60$, consistent with its $O(m^{-2})$ remainder. The exact pgf there is calculated using the coefficients of $\prod_{j=1}^{m-1} (1 + q^j)$, divided by 2^{m-1} , in (7.1).

At $y = p_{60}(60^3)$, the uncorrected Lambert chart has index error -1.96028 , while (6.4) has error $1.66805 \cdot 10^{-4}$. At $y = p_{18}(2^{18})$, the exact gamma-background inverse X has index error $2.99966 \cdot 10^{-4}$; adding the exponential correction in (6.8) reduces it to $-1.72742 \cdot 10^{-8}$. These diagnostics illustrate why algebraic and exponential inverse corrections should be treated separately.

8.4 Reproduction and package contents

The archive contains the article source and PDF, plus:

derive_coefficients.py

Exact Bernoulli/power-sum derivation of $L_r(c)$ and $b_r(c)$, with a selectable fixed order. The default is five.

asymptotics.py

Forward logarithmic profiles, the explicit cubic inverse, numerical inversion of a chosen truncated profile, and the first exponential inverse correction.

verify.py

Exact product and divisor-recurrence checks, finite bounds, OEIS initial-value checks, probability diagnostics, and inverse diagnostics.

Data files

`coefficients.json`, `exact_values.json`,
`critical_errors.csv`, `poisson_checks.csv`,
`inverse_checks.csv`, `verification_summary.json`.

Only SymPy and mpmath are required beyond the Python standard library. The article uses standard L^AT_EX packages and builds with three runs of `pdflatex`. The README supplies the exact commands. No network access is required to reproduce the included calculations after dependencies are installed.

[write] **Added 1 October 2026, batch 74: the shipped layout.** In the collection the four programs (with `build.sh`) are in `code/` and the data files, `requirements.txt` and the two run records in `data/`; this source is `article.tex`, renamed from `restricted_partitions_cubic_boundary.tex`, and the delivered PDF is not shipped. Every program reads and writes next to itself, so `verify.py` cannot find `coefficients.json` from `code/`, and `build.sh` still names the old source file. Rerun on a flat copy of `code/` and `data/`; the README gives the commands.

9 A formalization route from the existing Lean foundation

The exact combinatorial foundation already present in ProveIt suggests a clean separation into finite and analytic layers. This is a proposed route, not a claim of completed code.

The first layer would formalize the weak-composition orbit model and the transposition fixed-point injection. These provide an entirely discrete proof of the lower obstruction $m(m-1)^2/(4N)$. A companion lattice-simplex volume lemma gives the upper bound and the exact threshold. The existing multiplicity-vector equivalence can supply the bridge between the two models.

The second layer would formalize coefficient extraction for the finite q -Pochhammer product, the analytic identity (3.2), and the Bernoulli recurrence (3.4). At each fixed order, the explicit coefficient identities reduce to rational polynomial algebra and are good candidates for small proof certificates.

The genuinely analytic work is then localized: a uniform trigonometric index-count lemma, a Cauchy-circle decay bound, gamma-kernel integral estimates, and a uniform analytic Taylor remainder on a shrinking disk. The distinction between a fixed parameter and a constant uniform over $N \geq c_0 m^3$ must be represented explicitly in the quantifiers. Finally, probability is built from finite uniform measures on partition sets, coefficient-deletion identities, and factorial moments.

The source file inspected in the repository is compatible with this program, but neither its availability nor the successful Python checks certify any of these future Lean developments. In particular, a formalization must not replace the minor-arc estimate with the assertion that $q = 1$ is the only relevant singularity: the other roots of unity are present and must be bounded.

10 Further research questions and concrete next steps

The results above leave substantial, sharply formulated problems. The following are proposals for further work, not assertions that every one is historically open in exactly this form.

1. Optimal truncation and individual waves. Determine how the best order in (3.8) grows with m , and compute the leading nontrivial root-of-unity contribution in the cubic regime. The first

target is the $q = -1$ wave, followed by a uniform comparison of all remaining orders. This would turn the present fixed-order expansions into a substantially more complete transseries.

2. A uniform bridge below the cubic scale. Extend the centered analysis into $m^{5/2} \lesssim N \ll m^3$, where the centering factor can be large and the quadratic Bernoulli correction must itself be exponentiated. Match the resulting expansion to a small-parameter expansion of the Szekeres saddle. The precise validity range and matching error are more useful targets than simply recovering a leading logarithm.

3. Structural degree cancellation in logarithmic coefficients. The general construction only immediately gives $\deg D_j \leq 3j$, while D_1 and D_2 have degrees 3 and 5. Investigate a systematic sharper degree theorem, its leading coefficients, and a connected-object or cumulant interpretation. Such a result could reduce symbolic complexity and improve error estimates across many dilute sequences.

4. Higher probability corrections and quantitative convergence. Compute an all-order version of (7.4), then prove a uniform complex- u domain sufficient for extracting point probabilities with explicit error bounds. Obtain a total-variation estimate for the Poisson approximation, rather than inferring one from convergence of fixed moments.

5. Rare triple repetitions. The present proof bounds their probability by $O(1/m)$ at the cubic scale. Determine its leading constant and joint asymptotics with the number of doubletons. At other scales, identify when tripletons develop a nondegenerate limit and whether a hierarchy of multiplicity transitions results.

6. Deletions whose number grows. Lemma 7.1 assumes fixed k . Find the maximal growing range $k = k(m)$ for which a useful uniform coefficient ratio holds. This would give moderate-deviation information for Z_m , not just a Poisson limit at fixed counts.

7. Other weight systems. Replace $1, \dots, m$ by structured integer weights with an available unit weight and controlled power sums. The local expansion generalizes immediately, but the global arc estimate depends on the weights' arithmetic distribution. Find hypotheses that separate a universal volume/cumulant law from genuine periodic obstructions.

8. Certified inverse thresholds. Combine explicit finite constants in the remainder bounds with monotonicity to produce intervals enclosing the integer inverse $Q(y)$. A useful algorithm must report when its interval straddles two integers, rather than silently rounding a smooth chart. This is particularly important when y lies very close to an exact sequence value.

9. A proof-producing asymptotic pipeline. Starting from the finite-product generating function, automate power sums, Bernoulli coefficients, fixed-order remainder parameters, transseries substitution, and inverse reversion. The symbolic output should be a small certificate checkable independently, with numerical fitting excluded from the proof path.

10. OEIS and literature reconciliation. Prepare a narrowly scoped, source-backed update for the affected entries: replace the vague quartic-growth condition by a precise sufficient condition, attach the classical attribution, and separately record explicit refinements. A useful contribution is improved mathematical traceability, not relabeling classical consequences as breakthroughs.

11 Conclusion

The restricted-partition family admits one coherent explanation of several OEIS asymptotic patterns. The volume approximation has an exact cubic threshold, critical growth leaves a nontrivial constant, and larger sizes admit polynomial, exponential, or factorial-scale refinements governed by the same Bernoulli coefficient system. Uniform contour control is what turns formal local algebra into theorems about the whole counting sequence.

The inverse formulas illustrate a second important distinction: an exact gamma background is needed before exponentially small corrections can be meaningfully inverted. Finally, the conjugate partition model shows that the cubic boundary is not merely an artifact of asymptotic notation. It is a regime where repeated parts survive in a nondegenerate Poisson law, with a computable finite-size correction. The elementary criterion, all-order expansions, inverse charts, and probabilistic law form the substantive proof package supplied by this report.

A A compact all-order coefficient recipe

For a requested order K , the following is a finite exact procedure. Compute $S_{2r}(m)$ for $1 \leq r \leq K$ by Faulhaber’s formula or finite symbolic summation. Form γ_r , then a_j from (3.4). Let $x = 1/m$, $h = (x + x^2)/(4c)$, and form, modulo x^{K+1} ,

$$Q(x, c) = \sum_{j=0}^K a_j (1/x) \frac{x^{4j}}{c^{2j}} \prod_{i=1}^{2j} (1 - ix) (1 + h)^{-2j}.$$

Although the expression uses $a_j(1/x)$, the combined summand has no negative powers: its lowest possible degree is j . Then

$$\begin{aligned} \sum_{r=1}^K L_r(c) x^r &\equiv \log Q(x, c) + (x^{-1} - 1) \log(1 + h) - \frac{1}{4c} \\ &\quad - 2 \sum_{r \geq 1} \frac{B_{2r}}{2r(2r-1)} x^{2r-1} \pmod{x^{K+1}}. \end{aligned}$$

Only finitely many Stirling terms contribute. Apply (4.4) to obtain relative coefficients. This is the algorithm implemented in `derive_coefficients.py`; it uses exact rational arithmetic. For dilute coefficients, use (5.3), (5.2), and (5.5) instead. No numerical sequence values are needed to discover any of the displayed coefficients.

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